

Development of a Disability Eating Utensil Aid for Hand Tremors

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December 12th, 2025

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Executive Summary

Team “The Scoop” has identified a problem faced by people with hand tremors: eating with spoons. Spilling food because of a hand tremor can feel humiliating, so for those with hand tremors, being fed by another person is the only way to eat soup or other liquids. The team proposes to design a utensil that allows people with hand tremors to feed themselves. There are market solutions which dampen tremors with weight or mechanical action. Most are heavy, unsatisfactory, or expensive. The team will design a utensil that can be sold in large quantities for under \$50.

Current solutions fall into one of three categories described above: weighted handles, mechanical action, or electrical feedback. Weighted handles can be adequate for people with minor tremors but fail in two cases. The handle does not dampen severe tremors sufficiently, and many people with tremors lack muscle mass with which to hold the utensil. Mechanical action describes utensils which contain a counterbalanced pivot, like a gimbal. No design has been found which offers an acceptable range of motion or is easy to use. Electrical feedback utensils contain sensors and actuators which hold the spoon level. These products cost at least \$200 per spoon. Hospitals funded by Medicare treat hundreds of patients with tremors and cannot afford such an expenditure.

The team proposes to design a utensil that uses mechanical action to allow people with tremors to feed themselves. Upon meeting with the consumers twice, modifications were made to the original design to include devices that will ensure the utensil will not be dropped on the floor. The modifications that were suggested by the consumers will be taken into consideration moving forward with this design process.

Introduction

As of 2020, 24.91 million people globally experience hand tremors through some sort of neurological disorder, such as Parkinson's disease or multiple sclerosis (Song, 2021). Having hand tremors makes it difficult to eat using utensils and frustrates the person with hand tremors and the person who must feed them. This project aims to develop a low-cost mechanically stabilized utensil that reduces the effects of tremors while eating.

Following the Preliminary Design Review (PDR) and Critical Design Review (CDR), the team decided to add two areas on the utensil that ensure it does not fall out of their hands with wrist and hand straps.

This report describes an overview of the design and technical, cost, schedule, and risk analysis. The report also discussed the test and validation plans and change logs. The recommendations will be discussed at the end of the report.

Final Design Description

The objective is to create a utensil which dampens vibrations of the wrist and elbow in individuals who experience hand tremors. See table of the qualitative and quantitative requirements in Design Documentation 3.G.1.

The engineering requirements are as follows:

- Has a long and wide handle with an ergonomic grip 5.91 inches (150mm) in length, 1.38 inches (35mm) circular diameter
- Dishwasher Safe: must withstand at least 125 dishwasher cycles (Measurlabs, 2019)
- Be food-grade & FDA approved material. This means using 304 stainless steel for the dishwasher parts

The features and functions of the product are as follows, see Design Documentation 3.G.2:

- Scoop Food
 - Provide strong grip
 - Spoon-head geometry suited for common foods
- Reduce Spillage
 - Provides stability
 - Counter hand movements
 - Maintain leveling
 - Successfully deliver food to mouth
- Easy to Clean
 - Hand-wash safe
 - Dishwasher safe
 - Water resistant

- Corrosion-resistant

After analysis of the preliminary designs in the PDR stage, the team decided to move forward with a concept that revolves around using a bearing to stabilize the utensil head. This design satisfies all the design requirements mentioned; however, there are modifications to be made regarding the material to ensure it is hand-wash and dishwasher safe. For this, the team is planning on using stainless steel for the detachable portion that will need to be washed after each use.

Compliance Verification

The selected design option employs a bearing to stabilize the utensil. The utensil is attached to a bearing at one end with sufficient counterweighting to ensure no spilling occurs with radial rotation of the handle. This initial design concept (Instructables, 2023) allows for mechanical dampening of the motion of the utensil with minimal features (Design Documentation 3.I.1 right image). The current design has changed substantially from the initial design by adding multiple straps to the handle as well as a tab to restrict the rotational aid of the utensil, a feature deemed necessary after customer feedback. The utensil will be detachable, so that the rest of the design does not need to be dishwasher safe.

Final Technical Analysis

Technical analysis of the current design reveals several things the design does well and several things that do not meet the design standards. The current gimbal-like spoon has the following advantages and disadvantages:

Pros

- Successfully dampens movements because of shaking in the horizontal direction
- Has a tab which, when held, allows standard movement for the utensil
- Wrist strap and finger strap
- Washable parts are dishwasher safe

Cons

- Utensil mount connects to the bearing by press fit and may wear over time

The eating utensil design has reached its final stages and with moderate tweaks to its design and materials, it accomplishes all goals the team has set for it.

Test and Validation Results

The testing and validation plan for the product consists of four tests. The first test is a drop test. The drop test was from 2 meters high, which is the highest foreseeable height

the product will be used, and will ensure that if the user does drop the product, it will not break. The team tested this and found that the utensil survived and had no cracks or issues. Next, the product must withstand fourteen 1-hour dishwasher cycles. Running the detachable part of the product in the dishwasher fourteen times will ensure that it is dishwasher safe. The team tested this, and found that the product had no issues, and is dishwasher safe. Additionally, the utensil aid will undergo insertion and removal force tests for the interchangeable utensil heads. The insertion force that will need to be withstood is 7 lbf and the removal force is 3 lbf. These values were determined by understanding how much force is needed to open and close lids off Tupperware. The team tested this and found the following values for insertion and removal forces to be identical to that of Tupperware lids.

Change Log

The team has taken the initial prototype and met with a group of potential users for feedback multiple times. This feedback has led to numerous changes from the first design. Firstly, the rotating mechanism of the utensil made it impossible to scoop contents from the side of a bowl up, as the utensil would rotate to a vertical position as it contacted the bowl. To fix this, the team has added a tab in the form of the handle of the metal spoon that, with light pressure, will fix the utensil in place preventing the rotation of the device. This gives users the freedom to determine when the device helps stabilize and when it can be used as a regular spoon without the restrictions that come with a rotating device. The potential users noted several other related issues they have using a utensil that come along with their hand tremors. These are things such as suddenly freezing and dropping the utensil, losing grip strength during usage, or issues with the handle size. To fix these things, the team increased the handle size and added ribbed texture to help hold it. Other changes include two handle straps for the spoon, one around the fingers on the handle, and the other around the wrist. These straps can prevent the utensil and/or its contents from falling onto the floor if any abrupt freeze in the users' motion occurs or if they lose their grip strength during use.

Final Cost and Schedule

The 3-D print handle costs about \$2.50, stainless steel stock portion that was used costs about \$8, the cost for the utensil itself is roughly \$2, and the bearing was determined to cost \$1, making the entire disability eating utensil cost roughly \$13. This cost will be

reduced if production moves to a high scale as a lot of the parts can be bought in bulk for a much cheaper product, with the estimated cost to manufacture being \$4. This low cost enables the team to keep their low selling point of \$25 and potentially even lower that to become more appealing to potential customers. The team was unable to present the final product to their customers due to timing restrictions for the course; however, based on previous interactions, they believe that the feedback they would get would be very positive. The next steps for the team are to remain in contact with a professor from the business school to have a potential partnership in which this product design is patented or even launched into the market during the upcoming semester.

Risk Management/Mitigation

Safety of the user is the primary risk that the team will encounter during the project (Design Documentation 1.D.1). The eating utensil will be used to drink hot soup, and if it is not effective in terms of stabilization, the person may spill the food on themselves. With the target consumer being elderly, they may have weaker immune systems and other health complications. So, healing from a burn caused by spilling a hot item on themselves, for example, is something the team will make sure to mitigate.

The parts that will be washed will be made from food-grade metals and will contain no electric circuits. This will make them dishwasher safe and avoid risk of electrical shock. Risks are present while manufacturing and assembling the products as large machines will have to be operated. Such risks can be mitigated by proper training, supervision, and following shop safety rules.

One smaller risk that was overlooked initially is the potential for injury when picking up the utensil if it falls on the floor. To mitigate this, the team added a hand strap and wrist step to ensure that the utensil does not fall on the floor under any circumstances.

Final Recommendations

Following the preliminary design phase of the project, the team fell behind in customer feedback. The team met with potential users on October 1 and asked them to eat with the prototype utensil and a utensil with a weighted handle. The users were pleased with the prototype but asked for a way to attach the spoon to their hand or arm to prevent dropping it. Most of them were advanced in age, and bending over to pick up the utensil is difficult. The team has proposed a new strap like a Wii remote and is looking for handles that minimize the chance of dropping the utensil. The customers also needed a way to hold the bowl in place when scooping food, so the team proposed a tab which can be pressed against the bowl to prevent it from rotating.

In the October 20 meeting, some recommendations were put forward. For the eating utensil aid, it was suggested that the tab stopper be longer so that it is more accessible and having a more comfortable finger strap. Both recommendations were put into place in the final product design.

Following the completion of the design, the team did not get an opportunity to meet with the potential consumers again. Final recommendations to improve the design will be made solely based on the teams' ideas of a more convenient product for the consumers. This list will not be truly comprehensive until the product is on market, and potential issues with the product could be made clear. The first potential issue is the tab. The tab was simply removed, and the back end of the spoon was used instead. By applying a light amount of pressure to this part of the spoon, it prevents the rotation of the spoonhead, allowing the spoon to be used to scoop food or liquids from a bowl. This feature works just as well as the tab, and the force needed is no more than the tab required. The concern with this feature is that it is not initially clear; the team can see potential consumers accidentally holding this part, preventing the device from functioning as normal. Or alternatively never holding this part and being unable to scoop with the spoon. The team can only hope that this feature is made clear to consumers when the product is marketed in the future.

The next design change recommendation the team could think of was the ability for the device to be used with any household spoon. Currently the spoon needs to be drilled through and attached to the device by a screw, this means multiple spoonhead systems must be purchased by the user if the spoon is to be used multiple times without washing the spoonhead between each use. This could drive the cost upwards by up to \$11 for each spoonhead system. The spoon itself only costs \$2 but is connected to the steel piece via a screw. The team imagines it's not realistic to expect people with hand tremors to screw and unscrew new spoons between each usage of the product. In order to drive costs down, the team suggests a new method of connecting the spoon to the steel piece. The ideal connection method would allow any household spoon to be clicked into place onto the steel piece, although this may require a new connector piece as well. This type of connection would prevent the need to purchase custom spoons, driving the cost of the product down even further. Additionally, it would be more convenient for the consumers to use and wash than the current design.

Conclusion

The team has identified a problem; people suffering from hand tremors struggle to eat by themselves. The team has created a purely mechanical solution, a spoon which can stabilize itself by rotating as the hand trembles. The solution features a removeable spoonhead system composed of a bearing, steel connector piece, and the spoon itself.

When pressure is applied onto the backend of the spoon, the rotation of the device is prevented, allowing the spoon to scoop as a normal spoon would. The handle features a hand and wrist strap to prevent the spoon from falling out onto the ground due to hand freezes which sometimes may accompany tremors. One spoon system costs under \$25, well under the initial goal. Even after purchasing a few extra spoon head systems, the spoon remains under the initial cost goal of \$50 for a solution that works much better than the currently available affordable solutions. The design is simple and efficient, completing the teams' goal of being an affordable and effective solution.

cost approximately \$25, and the preliminary designs cost \$60. This leaves \$915 for any unexpected expenses and leaves the designs well under budget.

Design Documentation D: Risk Register

ID	Date Raised	Risk Details	Mitigation Plan	Owner
1	9/12/2025	There is a risk associated with running any machines within the mechanical engineering building at Purdue University.	The mitigation plan for this is to adhere to all requirements for each lab that is present in the access quizzes. The team will also have a professor or student worker for those labs overlook our process until both the team member and the staff are comfortable with the team member working alone.	Kaushal Patel
2	9/12/2025	There is a risk associated with the user being reliant on the disability eating utensil aid when it comes to holding a hot object.	The mitigation plan for this risk to do a lot of quality testing prior to releasing the product to the public. These tests will ensure that there will not be any spillage when someone tries to use the product.	Kaushal Patel
3	11/5/2025	There is a risk associated with the user picking up the utensil in the case that it falls on the floor while being used.	The mitigation plan for this risk is to have precautions in the form of a wrist and hand strap to ensure that the product stays connected to the users hand under any circumstances.	Kaushal Patel

Design Documentation 1.D.1: Risk Register

Design Documentation 2

Design Documentation E: Market Analysis

 Gyenno Utensils	 Bunmo Weight Utensils	 Nourislif Adaptive Utensils
• Very expensive • Excellent stabilization • Easy to swap out • App integrated • Dishwasher safe	• Inexpensive • Weighted handles may be hard to hold • No stabilization feature • Dishwasher safe	• Inexpensive • Larger grip (may be hard to hold for those with smaller hands, but texture handle helps) • No stabilization feature • Dishwasher safe

Design Documentation 2.E.1: Market Analysis of Utensils for Eating Disorders

Design Documentation F: Value Proposition

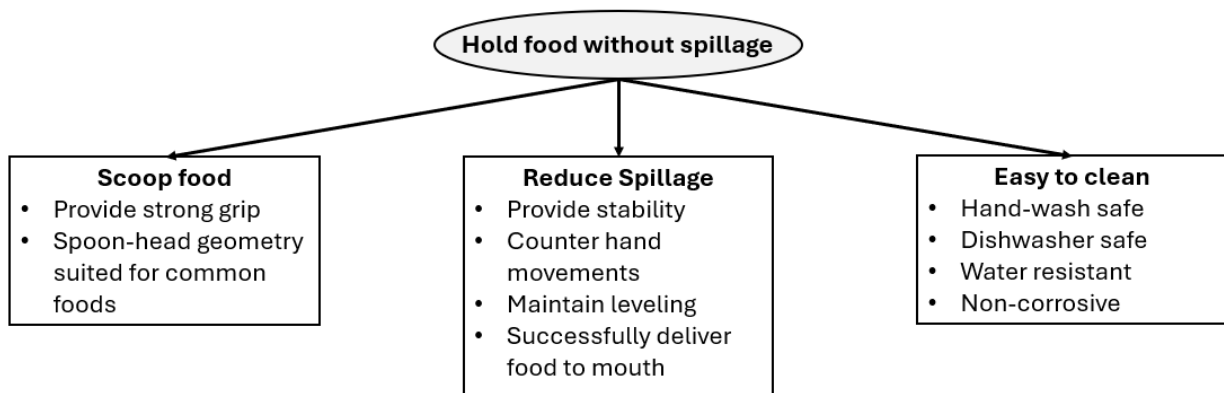
The team's product aims to be better than the competition by offering a solution which stabilizes the utensil across multiple axis of movement as well as being an affordable option for those affected. The product will also be fully mechanical which enables the user to clean the utensil easily and put it in the dishwasher. This means that the product will have no battery and electrical components. The team will make sure to market their product well in order to help those that could use the product be aware of its existence. They found that one of the retirement homes was not aware of options on amazon and Walmart, so the team will make sure to spread the word around of their product.

Design Documentation 3

Design Documentation G: Engineering Requirements & Constraints

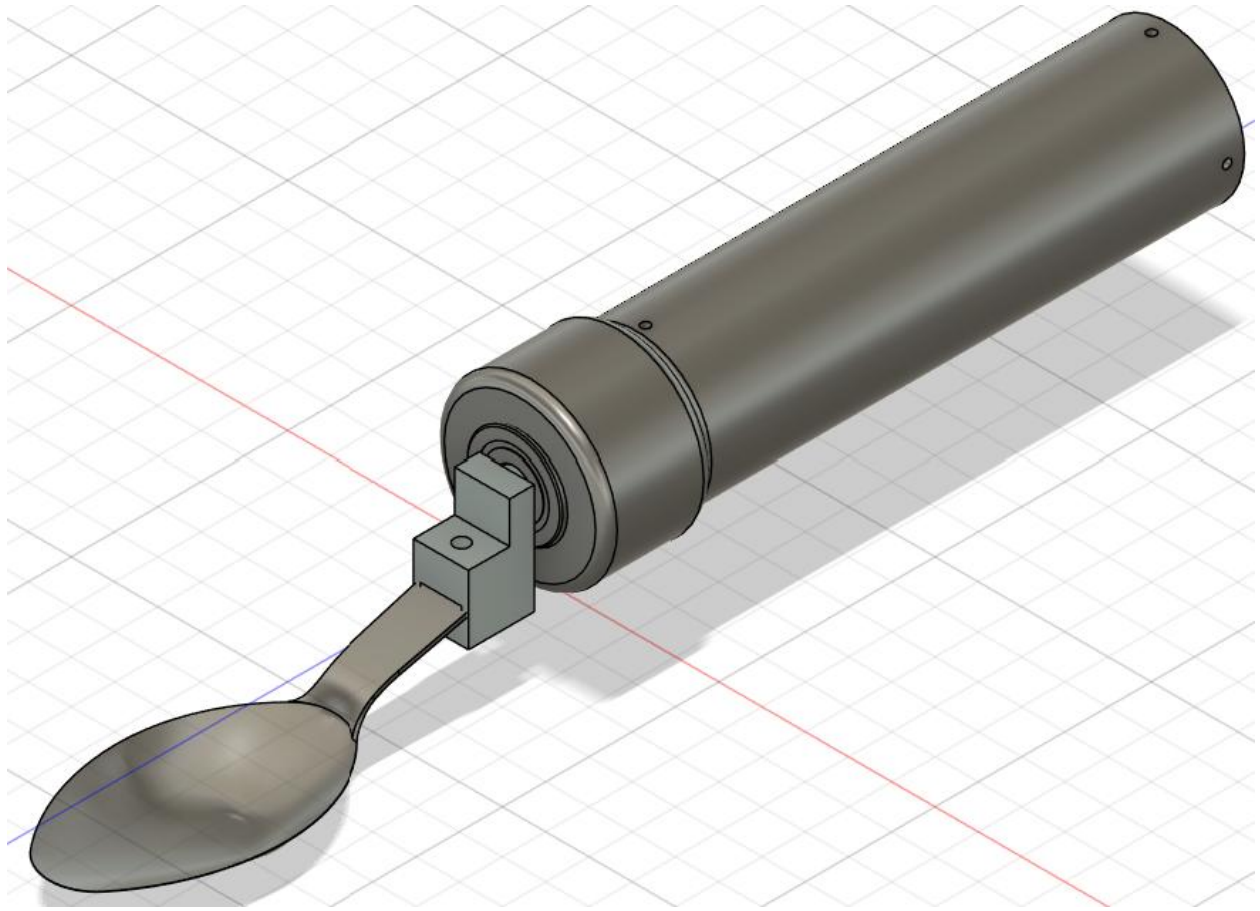
Qualitative Requirements	Quantitive Requirements
Long Handle	5.91 inches / 150mm (length)
Wide Handle	1.38 inches / 35mm (diameter)
Dishwasher Safe	125 Dishwasher cycles
Food Safe	304 stainless steel for dishwasher part

Design Documentation 3.G.1: Qualitative and Quantitative Requirements



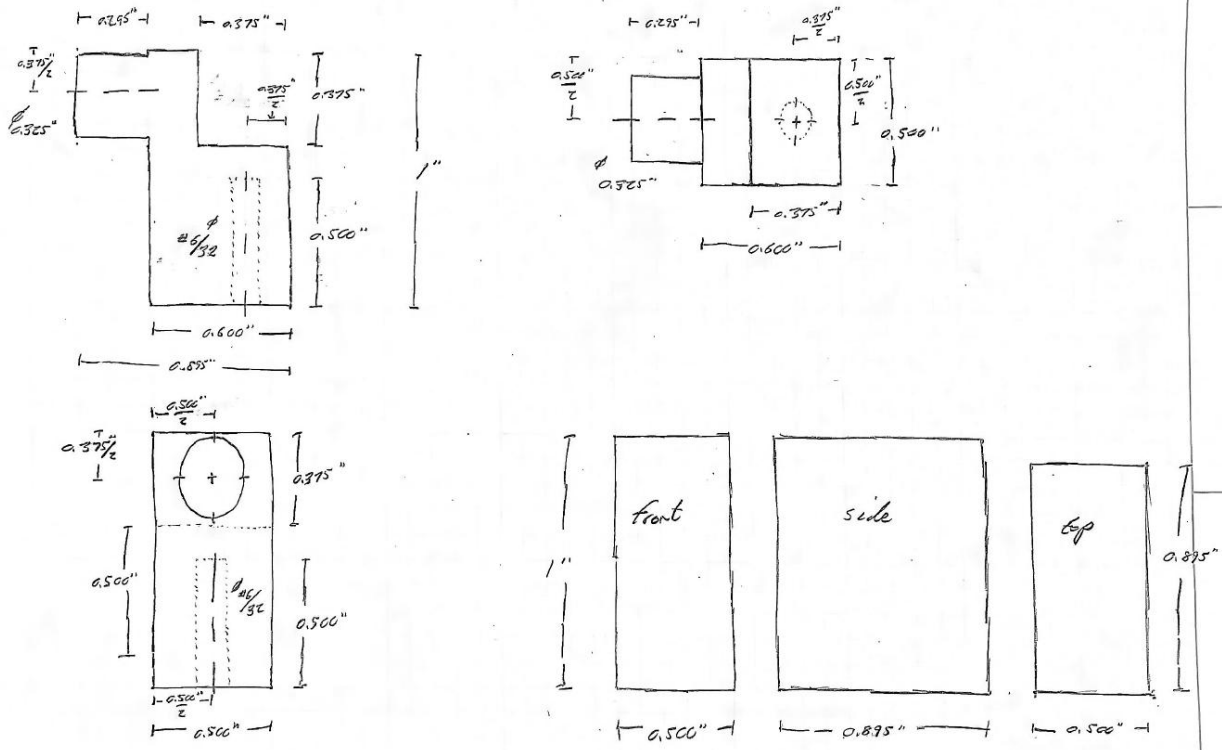
Design Documentation 3.G.2: Features and Functions

Design Documentation H: CAD



Design Documentation 3.H.1: CAD assembly of final design

Counterweight
by Christian Heide, 12/12/2025



Design Documentation 3.H.2: Hand drawn drawing of the final design of the stainless-steel connector

Design Documentation I: Analysis

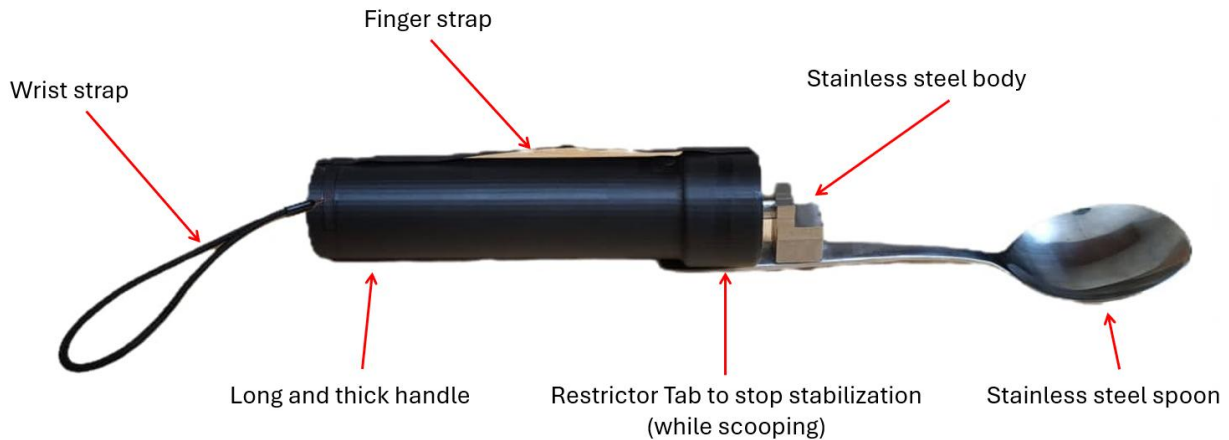


Weighted and Good Grip Handle



Self Stabilizing Spoon

Design Documentation 3.I.1: Down Selection Ideas



Design Documentation 3.1.2: Final eating utensil design assembly prototype

The final design of the eating utensil has a metal spoon head, and this has not increased the weight by much. For the handle, we are moving forward with the 3D printed material. A metal handle would not add any additional functionality, and it would be heavier to use. The finger strap is elastic so that people with all hand sizes can hold the handle comfortably. The parts connecting to the bearing and the spoon/fork head are stainless steel. The restrictor tab is now thicker and longer for ease of use.

Design Documentation J: FMEA

For the eating utensil, the team will conduct many quality tests to ensure the product matches what it sells. The first test will be a drop test onto carpet, wooden, and tile flooring. With the target audience being the elderly, it is important to make sure that the product can withstand being dropped on the floor by accident. The team will test the rotation and stabilization of the product. This will be done with objects of different weights at the end of the utensil head and conducted over weeks to ensure sure the utensil can perform this motion continuously.

Design Documentation K: BOM & Sourcing Plan

Item	Cost
PDR + CDR Costs	\$60
FDR Handle	\$5
FDR Stabilizing Parts	\$10
FDR Straw and Catcher	\$10
Marketing + Testing Plan + Unforeseen Circumstances	\$915

Design Documentation 3.K.1: Bill of Materials (BOM)

All parts will be sourced from Amazon, McMaster-Carr, Walmart, and Purdue ME resource rooms.

Design Documentation L: Validation Plan

Test	Metric
Drop Test	1-2 meters height
Insertion Force*	Within 3-7 lb-f
Removal Force*	Within 1-3 lb-f
Dishwasher Safe	14 1-hour dishwasher cycles

Design Documentation 3.L.1: Testing and validation plan

Insertion and removal forces will be tested using Instron tensile testing machine, and drop test will be done on carpet, wooden, and tile flooring.

The final prototype passed the drop test and dishwasher test. Moreover, based on qualitative analysis, the insertion and removal forces are low and similar to that of Tupperware lids.

References and Citations

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